

- Description

This module manages a smart solar panel cleaning system, ensuring that:

1. The system continuously monitors the dust accumulation level on the solar panel (expressed as a percentage).
2. If the dust level exceeds a predefined threshold (e.g., 50%), a cleaning cycle is automatically activated to remove dust from the panel.
3. If the dust level is below the threshold, no cleaning action is taken.

This system helps maintain optimal solar panel efficiency by ensuring that dust accumulation is minimized.

- Specification

module SmartSolarPanelCleaning

var currentDustLevel, dustThreshold: real

process updateDustLevel(newDustLevel: real) returns (currentDustLevel: real)

functional scenario form:

currentDustLevel > dustThreshold && output "ALERT: Dust level (" + currentDustLevel + "%) exceeds threshold (" + dustThreshold + "%). Cleaning cycle activated.", Cleaning cycle activated.

||

currentDustLevel <= dustThreshold && output "Dust level (" + currentDustLevel + "%) is within acceptable limits. No cleaning required.", No cleaning action required.

end process

end module

- Java Code

```
package code;
public class c108_SmartSolarPanelCleaner {
    private double dustThreshold;
    private double currentDustLevel;
    private boolean cleaningCycleActive;
    public c108_SmartSolarPanelCleaner(double dustThreshold, double initialDustLevel) {
        this.dustThreshold = dustThreshold;
        this.currentDustLevel = initialDustLevel;
        this.cleaningCycleActive = false;
    }
    public void updateDustLevel(double newDustLevel) {
        currentDustLevel = newDustLevel;
        adjustCleaningCycle();
        System.out.printf("Current Dust Level: %.2f%%\n", currentDustLevel);
        if (cleaningCycleActive) {
```

```

        System.out.println("ALERT: Dust level too high! Cleaning cycle activated.");
    } else {
        System.out.println("Dust level is within acceptable limits. No cleaning required.");
    }
}

private void adjustCleaningCycle() {
    if (currentDustLevel > dustThreshold) {
        cleaningCycleActive = true;
    } else {
        cleaningCycleActive = false;
    }
}

public double getCurrentDustLevel() {
    return currentDustLevel;
}

public boolean isCleaningCycleActive() {
    return cleaningCycleActive;
}

public static void main(String[] args) {
    c108_SmartSolarPanelCleaner cleaner = new c108_SmartSolarPanelCleaner(50.0, 40.0);

    // testcase-VT:
    // cleaner.updateDustLevel(45.0);
    // cleaner.updateDustLevel(55.0);
    // cleaner.updateDustLevel(48.0);

    // testcase-FT:
    // cleaner.updateDustLevel(1.2251194114176424E303);
    // cleaner.updateDustLevel(-1.7976931348623157E308);
    // cleaner.updateDustLevel(-1.515702054883285E308);

    // testcase-Z3:
    // cleaner.updateDustLevel(45.0);
    // cleaner.updateDustLevel(55.0);
    // cleaner.updateDustLevel(48.0);

    // testcase-UVT:
    cleaner.updateDustLevel(45.0);
    cleaner.updateDustLevel(55.0);
    cleaner.updateDustLevel(48.0);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c108_SmartSolarPanelCleaner.java	100.0 %	4	0	4
▼ c108_SmartSolarPanelCleaner	100.0 %	4	0	4
● updateDustLevel(double)	100.0 %	2	0	2
■ adjustCleaningCycle()	100.0 %	2	0	2
● c108_SmartSolarPanelCleaner(d		0	0	0
\$ main(String[])		0	0	0

3. Vibration testing instruction coverage: 92.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	70	21,915	21,985
▼ src	0.3 %	70	21,915	21,985
▼ code	0.3 %	70	21,915	21,985
▼ c108_SmartSolarPanelCleaner.java	92.1 %	70	6	76
▼ c108_SmartSolarPanelCleaner	92.1 %	70	6	76
● updateDustLevel(double)	100.0 %	28	0	28
■ adjustCleaningCycle()	100.0 %	14	0	14
● c108_SmartSolarPanelCleaner(d	100.0 %	12	0	12
\$ main(String[])	100.0 %	16	0	16

4. Fuzz testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c108_SmartSolarPanelCleaner.java	100.0 %	4	0	4
▼ c108_SmartSolarPanelCleaner	100.0 %	4	0	4
● updateDustLevel(double)	100.0 %	2	0	2
■ adjustCleaningCycle()	100.0 %	2	0	2
● c108_SmartSolarPanelCleaner(d		0	0	0
\$ main(String[])		0	0	0

5. Fuzz testing instruction coverage: 92.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	70	21,915	21,985
▼ src	0.3 %	70	21,915	21,985
▼ code	0.3 %	70	21,915	21,985
▼ c108_SmartSolarPanelCleaner.java	92.1 %	70	6	76
▼ c108_SmartSolarPanelCleaner	92.1 %	70	6	76
● updateDustLevel(double)	100.0 %	28	0	28
■ adjustCleaningCycle()	100.0 %	14	0	14
● c108_SmartSolarPanelCleaner(d	100.0 %	12	0	12
\$ main(String[])	100.0 %	16	0	16

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c108_SmartSolarPanelCleaner.java	100.0 %	4	0	4
▼ c108_SmartSolarPanelCleaner	100.0 %	4	0	4
● updateDustLevel(double)	100.0 %	2	0	2
■ adjustCleaningCycle()	100.0 %	2	0	2
● c108_SmartSolarPanelCleaner(d		0	0	0
\$ main(String[])		0	0	0

7. Z3 instruction coverage: 92.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	70	22,959	23,029
▼ src	0.3 %	70	22,959	23,029
▼ code	0.3 %	70	22,959	23,029
▼ c108_SmartSolarPanelCleaner.java	92.1 %	70	6	76
▼ c108_SmartSolarPanelCleaner	92.1 %	70	6	76
main(String[])	100.0 %	16	0	16
c108_SmartSolarPanelCleaner(double)	100.0 %	12	0	12
adjustCleaningCycle()	100.0 %	14	0	14
updateDustLevel(double)	100.0 %	28	0	28

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c108_SmartSolarPanelCleaner.java	100.0 %	4	0	4
▼ c108_SmartSolarPanelCleaner	100.0 %	4	0	4
updateDustLevel(double)	100.0 %	2	0	2
adjustCleaningCycle()	100.0 %	2	0	2
c108_SmartSolarPanelCleaner(double)		0	0	0
main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 92.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	70	22,959	23,029
▼ src	0.3 %	70	22,959	23,029
▼ code	0.3 %	70	22,959	23,029
▼ c108_SmartSolarPanelCleaner.java	92.1 %	70	6	76
▼ c108_SmartSolarPanelCleaner	92.1 %	70	6	76
main(String[])	100.0 %	16	0	16
c108_SmartSolarPanelCleaner(double)	100.0 %	12	0	12
adjustCleaningCycle()	100.0 %	14	0	14
updateDustLevel(double)	100.0 %	28	0	28